

# Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

**Golden rule:** nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py` .

## Status legend

|               | Meaning                                                                          |
|---------------|----------------------------------------------------------------------------------|
| ✔ Validated   | Survived friction-aware, regime-split backtest (still preliminary — see caveats) |
| ● Marginal    | Real effect, but fragile net of costs / not sign-stable                          |
| ✖ Failed      | Tested and does not work (kept here so we don't retry it)                        |
| 🧪 Untested    | Built and wired, awaiting data to evaluate                                       |
| 💡 Idea        | Backlog — not yet built                                                          |
| 🔄 In progress | Data/build task underway                                                         |

## Progress at a glance

Last updated: 2026-06-02. Update this section whenever a status, metric, or next-step changes.

## Strategy scoreboard

| Strategy                                                                      | Class         | Status                                                  | Headline result                                                                                                                                                                                                                                                                                     | Next action                                                                                                                |
|-------------------------------------------------------------------------------|---------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>retail_fear</b> (capitulation / bear-extreme / pessimism-fade)             | Macro overlay | ✓ Validated                                             | <b>excess vs buy-&amp;hold</b> , generalises across a risk-on basket: Gold +0.6 / AUDJPY +0.5 (win 88%) / SP500 +0.2pp robust; Silver +2.2pp lumpy; VIX gate ~4×                                                                                                                                    | live via MT5<br><b>RedditMacro_EA</b> ; vol-target sizing; longer history                                                  |
| <b>rrai_momentum_long</b> , <b>froth_high_long</b>                            | Macro overlay | ✗ No edge                                               | positive net but ~0 <b>excess vs buy-&amp;hold</b> (just bull drift)                                                                                                                                                                                                                                | dropped                                                                                                                    |
| <b>rrai_euphoria_short</b> + all macro shorts                                 | Macro overlay | ✗ Failed                                                | −1.5 to −3pp excess (shorting the bull)                                                                                                                                                                                                                                                             | dropped                                                                                                                    |
| <b>dump_fade_DELETION_short</b>                                               | Equity        | ✓ Thesis validated (forensic) / ⚠ not yet PIT-tradeable | <b>gone≥0.20 → +14.3%/10d short, 83% win, t=2.57, +ve in all 3 regimes</b> ; control proves the gate flips the trade (un-gated pump short <i>loses</i> −10.8%, t=−2.22)                                                                                                                             | make tradeable: finish author sweep to power the PIT<br><b>young_frac</b> gate (now n=3–7, inconclusive); express via puts |
| <b>distribution_short</b><br>(concentrated pump + sentiment–price divergence) | Equity        | ● Promising — <b>PIT-tradeable</b>                      | concentrated pump (per_author≥2, m≥10) + sentiment≥0.4 + 5d price already rolling over → <b>+7.2%/10d net short, t=2.70; date-clustered t=2.52 (47 indep. dates), dose-response in sentiment, +ve all 3 sub-periods, survives 2× costs</b> ; control (same pumps, no divergence) <i>loses</i> −4.1% | OOS-collect forward; tighten <b>per_author</b> (some ETF/large-cap leak in); express via puts; promote on more episodes    |
| <b>fade_organic_short</b>                                                     | Equity        | ● Marginal                                              | +0.83%/5d but P3 flips                                                                                                                                                                                                                                                                              | retry via puts + VIX/vol gate                                                                                              |
| <b>pump_long</b> (ride)                                                       | Equity        | ✗ Failed                                                | +3% but regime-flips, t=0.75                                                                                                                                                                                                                                                                        | — drop                                                                                                                     |
| <b>dump_fade_short</b> (naive)                                                | Equity        | ✗ Failed                                                | −9.9% (squeezed)                                                                                                                                                                                                                                                                                    | superseded by deletion-gated                                                                                               |
| <b>organic_long</b>                                                           | Equity        | ✗ Failed                                                | −4.2%, sign-stable loser                                                                                                                                                                                                                                                                            | inverse → <b>fade_organic_short</b>                                                                                        |

## Build & data progress

| Component                                                                         | Status                           | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature store ( <code>feature_daily</code> / <code>aggregate_daily</code> )       | ✓ Done                           | ~77k ticker-day rows, point-in-time                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Friction-aware backtester ( <code>backtest/engine.py</code> )                     | ✓ Done                           | costs + regime split + no-lookahead                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>/signals</code> feed (multi-strategy, json mt5)                             | ✓ Done                           | strategy-tagged; live, currently <b>flat</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Macro strategy R&D ( <code>backtest/macro_research.py</code> )                    | ✓ Done                           | only <code>retail_fear</code> beats buy-&-hold; momentum/froth no edge                                                                                                                                                                                                                                                                                                                                                                                                                          |
| MT5 framework ( <code>SignalClient.mqh</code> + <code>RedditMacro_EA.mq5</code> ) | ✓ Done                           | strategy-selectable thin client; demo-only                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Author profiling coverage                                                         | 📊 20%                            | 30,015 / 152,040; 2,973 gone (2,087 del / 886 susp); target ~40%                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Deletion-gated short testability                                                  | ✓ Forensically validated         | gone ≥ 0.20 short +14.3%/10d, t=2.57 (n=30); PIT <code>young_frac</code> proxy still sparse (n=3-7)                                                                                                                                                                                                                                                                                                                                                                                             |
| Short-interest / squeeze data (FINRA Reg SHO)                                     | ✓ Done                           | <code>scraper/finra.py</code> : 475k rows / 5,693 tickers; <b>z-score</b> screen (abs level is MM noise, median 0.49)                                                                                                                                                                                                                                                                                                                                                                           |
| Independent fear-confirmation gate (Wikipedia)                                    | 🟡 → 🟢<br>Strengthened on 2016-26 | <b>Extended series (525 caps, <code>backtest/fear_gate.py</code> ):</b> <b>fear-only</b> excess SPY +0.57 (day t=1.64, <b>episode t=2.77</b> ) / QQQ +0.80 (day t=1.91, <b>episode t=2.37</b> ) — effect <i>grew</i> with the 2018 episodes & SPY/QQQ now <b>agree</b> ⇒ likely real (day-level still marginal). The LIVE <b>trend+fear still ≈0</b> (SPY +0.10/QQQ +0.11) → <b>trend filter should be dropped, use fear-only</b> . AUDJPY stays ungated (fear <i>hurts</i> it, +0.21 → +0.09). |
| Live screens in <code>/signals</code> ( <code>tradeable:false</code> )            | ✓ Done                           | <code>smallcap_pump_fade</code> · <code>fresh_pump_alert</code> · <code>coordination_flag</code> · <code>biotech_catalyst</code> · <code>attention_fade</code> · <code>calendar_overlay</code>                                                                                                                                                                                                                                                                                                  |
| Idea backlog                                                                      | 💡 7 ideas                        | squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination                                                                                                                                                                                                                                                                                                                                                                                                    |
| Equity/options paper client (Alpaca)                                              | 💡 Planned                        | not started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data  $\leq D$ ; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.
6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

## Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Index / risk-FX** — *macro overlay only* (retail positioning as a contrarian factor).
- **FX majors / "Reddit sentiment on SPY as direction"** — no edge; don't build it.

# 1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

**Macro R&D verdict ( `backtest/macro_research.py` , **excess over buy-and-hold** ).** A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had ~0 excess (pure drift); all shorts were -1.5 to -3pp (fighting the trend). **One real macro edge, three formulations of it.**

## ✓ `retail_fear` — buy risk when retail capitulates (*served live*)

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) → contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct ≤ 0.15` **AND** `VIX ≥ 18` ; hold ~10d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d): excess over buy-&-hold** SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate ~4x'd it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Instrument scan (17 MT5 instruments, 10d excess vs buy-&-hold):** the signal **generalises across the risk-on basket** — Silver +2.22pp (lumpy, one big rally → size small), Gold +0.59pp (robust, +ve every regime), AUDJPY +0.47pp (steadiest, win 88%), SP500 +0.17pp. Oil (+1.82pp but one-episode) and crypto (BTC net-negative) **dropped**. Generalisation across assets is itself evidence it's a real risk-appetite signal.
- **Independent fear-confirmation gate (Wikipedia, LIVE — but now DOUBTFUL):** the **equity** legs (US500/USTEC) additionally require an *independent* fear spike — `fear_z ≥ 0.5` , the z-score of Wikipedia fear-page attention (Stock\_market\_crash / Recession / Bear\_market views) vs the trailing ~30d — **AND** a 200-DMA uptrend. Original rationale: SPY capitulation *alone* doesn't beat buy-and-hold (master backtest excess -0.05); an *ad-hoc* run reported **+0.36pp/10d when `fear_z` confirms vs ~-0.01 ungated**, and fear-attention is **1.32x higher on capitulation days** ( `DATA_PLAN.md` ). Wired live in `analytics/signals.py` ( `_fear_z` , `FEAR_MIN` ; `fear_z` in `/signals` ).
- **MT5 Strategy-Tester A/B (US500, D1, 2020–2026, long-only, real spread/swap; `mq15/RedditMacro_US500*.ini` , dates via `python main.py --export-signals` ).** This tests the gate as an **absolute long-only rule** (NOT excess-vs-buy-&-hold):

| variant                               | signals | trades | win% | net \$       | PF   | maxDD% |
|---------------------------------------|---------|--------|------|--------------|------|--------|
| baseline (all caps)                   | 335     | 100    | 57.0 | <b>+12</b>   | 1.00 | 13.6   |
| <code>fear_z ≥ 0.5</code> only        | 95      | 45     | 53.3 | <b>-1045</b> | 0.68 | 13.0   |
| trend up (200-DMA) only               | 223     | 71     | 56.3 | <b>+21</b>   | 1.01 | 9.0    |
| trend <b>AND</b> fear (the live gate) | 54      | 27     | 55.6 | <b>-372</b>  | 0.73 | 7.1    |

**Result: the `fear_z` gate does NOT help — it hurts.** Adding fear makes US500 *worse* both alone (-\$1045) and with trend (-\$372 vs trend-only +\$21). Fear spikes cluster *at/just before tops* where the 200-DMA still reads up (e.g. Feb-2020) → **trend AND fear** fires **long into the COVID falling knife**. The **trend filter** is the only component that helps, and mostly by cutting drawdown (13.6→7.1%), not return. Baseline ≈ breakeven re-confirms equity capitulation has no absolute edge. - ✓

**Committed EXCESS study ( `backtest/fear_gate.py` , `python -m backtest.fear_gate` ) — **this settles it.** The frame MT5 can't measure: mean EXCESS over buy-&-hold (the +0.36pp's own metric). Same point-in-time date-sets as the MT5 run (shared `capitulation_sets()` ). 10d, net of costs:**

| leg              | variant           | n   | excess%      | win% | t(exc)       |
|------------------|-------------------|-----|--------------|------|--------------|
| SPY              | baseline          | 332 | -0.05        | 61   | -0.21        |
| SPY              | <b>fear</b>       | 94  | <b>+0.32</b> | 68   | +0.64        |
| SPY              | trend             | 220 | -0.03        | 60   | -0.16        |
| SPY              | trend+fear (live) | 53  | +0.09        | 64   | +0.19        |
| QQQ              | baseline          | 332 | +0.06        | 62   | +0.20        |
| QQQ              | <b>fear</b>       | 94  | <b>+0.41</b> | 65   | +0.71        |
| QQQ              | trend+fear (live) | 53  | <b>-0.20</b> | 60   | -0.33        |
| AUDJPY (ungated) | baseline          | 333 | <b>+0.26</b> | 68   | <b>+2.04</b> |

**Verdict (three honest parts):** (1) **The +0.36pp REPRODUCES as a point estimate** — fear-only SPY +0.32 / QQQ +0.41 excess — so it was *not* fabricated; the fear gate does lift equity-capitulation excess. (2) **But it is NOT statistically significant** (excess  $t \approx 0.6$ ,  $n \approx 94$ ) — a thin, unproven edge. (3) **The LIVE trend+fear combo is the wrong way to apply it:** the trend filter *removes the buy-fear-in-a-downtrend bounces that carry the excess*, diluting SPY (+0.32  $\rightarrow$  +0.09) and flipping QQQ negative (+0.41  $\rightarrow$  -0.20). - **Reconciliation (why MT5 said "loss" and this says "small win"):** MT5 sums *absolute* P&L and holds one position at a time, so a few COVID-cluster trades dominate the dollar total (-\$1045); the excess frame means-averages independent trades, where fear is mildly +ve. Both are correct for their question — for *signal research* the excess frame is the right one, so the earlier "fear-gate is a drag" was a frame artifact. - **The genuine tension:** fear  $\rightarrow$  better *timing/excess* but worse *drawdown* (COVID knives); trend  $\rightarrow$  better *drawdown* but worse *excess*. The live **trend AND fear** gets the worst of both for excess. - **Recommendation:** keep the gate 🟡 (positive but insignificant). If we ever optimise the equity legs for excess, use **fear-only, not trend+fear**; AUDJPY (the one significant edge, +0.26  $t=2.04$ ) stays **ungated** — confirmed correct. (User chose to leave the live gate unchanged for now; this is logged for when we revisit.) - 📊 **Power / effective sample (day vs episode — the divergence IS the verdict):** the 94 fear-days are *not* 94 independent obs — overlapping 10d windows + episode clusters. Treated as **days**, fear-only is  $t=0.64$  (SPY)/0.71 (QQQ); collapsed to **independent episodes** (~40, one vote each) it's  **$t=2.28$  (SPY)/1.85 (QQQ)**. The gap is driven almost entirely by the **COVID-2020 cluster** (~10+ losing fear-days  $\rightarrow$  one episode vote), so the effect is *positive across most episodes but fragile* — and SPY (2.28) vs QQQ (1.85) disagreeing on near-identical instruments over the same episodes says we're at the **noise floor**, not a stable edge. **Data to settle it:** day-level needs ~9× more obs for  $p < 0.05$  (~19× for 80% power) — impractical (~40+ yrs at ~15-20 fear-days/yr); the real lever is **independent fear episodes**, and the cheapest source is *backward* (Wikipedia fear data already reaches 2015) — **extending the RRAI/capitulation series to 2016-2017 adds the 2018 Volmageddon + Q4-2018 episodes** (price coverage starts 2016-05). - ✅ **DONE — RRAI extended to 2016-06 (525 caps, +190)**. Re-ran the study on the longer series, and the fear-gate **strengthened** (effect *grew* out-of-sample — the hallmark of a real edge):

| leg | fear excess (94d $\rightarrow$ 141d) | day t ( $\rightarrow$ ) | episode t ( $\rightarrow$ )    |
|-----|--------------------------------------|-------------------------|--------------------------------|
| SPY | +0.32 $\rightarrow$ <b>+0.57</b>     | 0.64 $\rightarrow$ 1.64 | 2.28 $\rightarrow$ <b>2.77</b> |
| QQQ | +0.41 $\rightarrow$ <b>+0.80</b>     | 0.71 $\rightarrow$ 1.91 | 1.85 $\rightarrow$ <b>2.37</b> |

**Both equity legs are now episode-level significant ( $t \approx 2.4-2.8$ ) AND agree** — the earlier SPY $\neq$ QQQ noise-floor red flag is gone. Day-level is still only marginal (1.6-1.9), so not fully nailed, but this is a real upgrade from "fragile/unproven." The trend filter conclusion **hardens:** **trend+fear** (live) is still  $\approx 0$  excess (SPY +0.10/QQQ +0.11)  $\Rightarrow$  **drop trend, use fear-only** on the equity legs. AUDJPY confirmed best ungated (baseline +0.21  $t=2.00$ ; fear-gating *hurts*  $\rightarrow$  +0.09). - **Expression:** a **diversified risk-on basket** — AUDJPY + Gold (XAUUSD) + US500/USTEC + small Silver (XAGUSD). Served via `/signals` (tag `retail_fear`); traded by `mql5/RedditMacro_EA.mq5` (demo). Equity legs gated by `trend + fear_z`. - **Caveats:** single ~12mo bull regime; **46 cap-days = only ~13 distinct fear episodes** (clustered  $\rightarrow$  effective  $n$  small); overlapping windows; edge small in absolute terms. - **Next:** leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

## ✗ Macro candidates that failed (kept so we don't retry)

- `rrai_euphoria_short`, `bull_extreme_short`, `froth_high_short`, all momentum-shorts: **-1.5 to -3pp excess** — shorting a bull market.

- `rrai_up_momentum_long`, `froth_high_long` : ~0 excess — just drift, no signal.

## 2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

### ✗ `pump_long` — ride concentrated promotion

- **Thesis:** concentrated mention spike (few authors, high volume) pops short-term.
- **Result:** net +3%/5d but **sign-flips** (P1 +15.8 → P3 -4.6), win 36%,  $t=0.75$ . The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust.**

### ✗ `dump_fade_short` — short a decelerating pump (naive)

- **Result:** net -9.9% (P1 -29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

### ✗ `organic_long` — buy broad attention spikes

- **Result:** sign-stable -4.2% net ( $t=-4.25$ ,  $n=321$ ). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

### ● `fade_organic_short` — fade broad attention spikes

- **Thesis:** broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result:** net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal.** Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.

### ✓ `dump_fade_DELETION_short` — short a pump whose promoters vanished (*THE thesis — forensically validated*)

- **Thesis:** concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. The original A2 idea, and the *reason* we track account lifecycle.
- **Result (2026-06-02, 20% author coverage,  $h=10$ , market-neutral, 75bps + 8bps/day borrow):** | gate | n | net short | win | t | regimes | |---|---|---|---|---| | `gone_frac` ≥ 0.10 | 33 | +10.0% | 81.8% | 1.51 | -1.2 / +19.9 / +11.4 | | `gone_frac` ≥ 0.20 | 30 | +14.3% | 83.3% | 2.57 | +11.4 / +27.7 / +3.9 | | **control:** pumps w/ promoters *still here* ( `gone`=0 ) | 202 | -10.8% | 36.1% | -2.22 | -30.7 / -0.6 / -1.1 |
- The gate **flips the sign of the trade** (~+25pp spread). Un-gated concentrated pumps are *un-shortable* (they continue / squeeze); only the ones whose promoters vanish collapse. The deletion is doing the work, not pump-selection. Positive across all 3 regimes at `gone` ≥ 0.20.
- ⚠️ **Not yet point-in-time tradeable.** `gone_frac` uses *hindsight*: Reddit exposes no deletion timestamp, so "gone" means "gone as of our latest sweep" — unknowable at the pump date. The PIT-available leading proxy is `young_frac` (account age at post time), but it's currently too sparse to conclude ( $n=3-7$ , and the few events lean the wrong way).
- **Path to tradeable:** (1) **finish the author sweep** (20%→40%+) to power the `young_frac` PIT test; (2) failing that, a **live deletion-monitor** rule — `author_status_revalidate` already detects deletions forward; short on deletion-*confirmation* within K days of a fresh pump, capturing the back half of the collapse. Express via **puts** (defined risk vs squeeze tails; many are HTB/no-borrow).

### ● `distribution_short` — concentrated pump + sentiment-price divergence (*NEW, point-in-time tradeable*)

- **Thesis:** the "bagholder distribution" moment — a *manufactured* pump (concentrated chatter) where **price has started rolling over but the crowd is still euphoric**. Insiders distribute into the retail bid; sentiment lags price. Every input ( `per_author` ,

mentions, sentiment, trailing price return) is known at signal time → **unlike the deletion gate, this is actually tradeable live.** ( `backtest/microstructure.py` )

- **Rule:** `per_author≥2 AND mentions≥10 AND sentiment≥0.4 AND 5d-trailing-return<0` → SHORT.  $h=10$ , market-neutral, 75bps + 8bps/day borrow.
- **Result (2026-06-02):** **+7.2%/10d net,  $t=2.70$ , win 52%,  $n=60$**  across **39 distinct tickers**.
- **Date-clustered test** (1 obs/date, kills intra-day correlation): **+7.6%,  $t=2.52$ , 47 independent dates** — *not* a risk-off-day artifact (max 3 events on any date, spread over 12 months).
- **Dose-response** in sentiment ( $s \geq 0.3/0.4/0.5 \rightarrow +3.9/+7.2/+11.1\%$ ) and **monotone in horizon** ( $h=5/10/20 \rightarrow +5.6/+7.2/+9.5\%$ ,  $t$  rising to 2.92) — the hallmarks of a real effect, not a fit.
- **Survives 2x costs** (150bps spread + 16bps/day borrow → still +4.9%/10d).
- **Control:** same concentrated pumps *without* the divergence (price up) → **-4.1%** — the divergence is doing the work. Positive in all 3 sub-periods.
- **Why it works where the others don't:** the gate is *precise* — it only fires inside the manufactured subset. The broad high-attention universe **drifts flat-to-up** (gross  $\approx +0.04\%$ ), so generic pump-shorting just pays costs (see negatives below); the edge requires isolating the distributing pumps, which both this and the deletion gate do.
- **Caveats / next:**  $n$  still modest (47 indep. dates); single  $\sim 12$ -mo window (sub-periods, not true bull/bear regimes); small-cap-heavy so no multi-regime price history; loose `per_author≥2` lets some ETFs/large-caps (QQQ/IWM) leak in — tighten to  $\geq 3$  and/or split small-cap vs index. **Collect forward OOS** (the clock starts now); express via **puts**. Not yet promoted to the live `/signals` feed.

### ✗ Negatives from the same study (recorded so we don't re-run them)

- **Broad-universe microstructure timing does NOT survive costs.** Recruitment-exhaustion, mention-deceleration, and sentiment-price divergence applied to the *whole* high-attention set (`mentions_z≥1`) are all **net-negative** — the universe drifts flat-to-up, so shorting it generically just bleeds the  $\sim 2.3\%$  friction. Precision (concentration filter) is mandatory.
- **Recruitment-exhaustion alone = null.** New-author inflow rolling over is directionally supportive *within* concentrated pumps (+1.8% net, but  $t=0.33$ ) and adds to the stack, but is not a standalone edge at current coverage. The new-author feature is built and kept.
- **Up-leg longs both fail.** (a) "Ride while recruiting + accelerating" (concentrated) netted +1.6% but win 27%, regimes [+21/-7/-9] — one big winner, otherwise negative. (b) "Attention-leads-price coil" (broad: recruitment surging while price still flat) was **net -1.4 to -1.9%** across horizons, sign-stable loser. Neither is a tradeable long; dropped.

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### 3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) + **short pressure** → squeezable. *Partially built:* `scraper/finra.py` + `squeeze_screen()` flag tickers with an anomalous **z-score jump** in daily short-ratio vs their own baseline (HTZ/GME currently surface). *Still needs* true SI%/float/days-to-cover (FINRA **bi-monthly short-interest** report) + borrow-fee for the level; the daily file's absolute level is market-maker noise. *Not yet backtested.*
  - **Options-flow signal:** parse contract-level chatter (unusual\_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
  - **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
  - **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
  - **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
  - **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
  - **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.
-



## 4. Data gaps blocking strategies

| Gap                                           | Unblocks                                                                                                                                                      |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intraday prices</b> (have EOD only)        | timing pump/dump entries/exits (they move intraday)                                                                                                           |
| <b>Short interest / float / borrow fee</b>    | squeeze <i>level</i> (FINRA daily short-volume now in via <code>scraper/finra.py</code> , but it's MM noise at the level; need bi-monthly SI%/float + borrow) |
| <b>Options chains / IV</b>                    | options expression; options-flow signal                                                                                                                       |
| <b>Longer history</b> (data-limited to ~12mo) | true multi-regime validation of the RRAI overlay                                                                                                              |
| <b>Author coverage</b> (20% → target ~40%+)   | a <i>point-in-time tradeable</i> deletion short — thesis already forensically validated; PIT <code>young_frac</code> proxy needs more coverage                |

## 5. Execution stack

- `/signals feed` ( `analytics/signals.py`, GET `/signals json|mt5` ) — live, serves the capitulation overlay. ✓
- **MT5 EA** ( `mq15/RedditRRAI_EA.mq5` ) — thin client for the index/FX overlay; demo-only. ✓
- **Alpaca / IBKR paper client** — for the equity/options legs. 🕒 planned.

## ★ MULTI-REGIME VALIDATION (2020-2026, incl. COVID crash + 2022 bear)

The deep-history backfill let us test across regimes — the result that actually moves confidence: - **AUDJPY capitulation-long is REGIME-ROBUST** — positive in EVERY year 2020-2026, including the 2022 bear (+0.7%, win 60%), win rates 60-93%. Graduates from preliminary → **validated**. This is the core tradeable edge. - **Equity (SPY) capitulation-long FAILS in a sustained bear** (2022: -0.7%/10d, win 43%) — falling-knife problem → **must gate with the 200-DMA trend filter** (don't buy fear in a downtrend). Confirmed needed. - **Euphoria-short only works in the bear** (2022 +1.3%; loses every bull year) → shorts are **regime-conditional**, switch on only in confirmed downtrends. (Vindicates not building shorts earlier.) - Caveats: small n/year (18-60), overlapping windows, historical RRAI is a WSB-heavy sampled backfill.

## Session findings — sector / calendar / latency (2026-06-01)

- **Sector segmentation rescues the pump signal** ( `backtest/sectors.py` ). The broad pump was a wash because it *mixed opposites*: **small-cap concentrated pumps FADE** (-14.6%/10d excess, win 22%,  $t = -3.93$  — pump-and-dumps reliably reverse → a short/avoid), while **biotech pumps are a fat-tailed lottery** (+66% mean, win 48%,  $t = 2.23$  — a few catalyst moonshots). Thematic universes (ai/quantum/space/nuclear/meme/frontier) had too few concentrated-pump events to judge.
- **Pump detection is TOO SLOW** ( `feature_daily` ). Median `novelty_days` at flag = **152**; only **1%** of pumps flagged within 3 days. Root cause: the `min_total ≥ 20` universe filter + trailing z-windows mean a ticker must accumulate ~20 mentions before it's even eligible → we catch *established-ticker spikes*, not fresh pump-and-dumps. Only **33 fresh day-0 bursts** exist in the data. **Fix**: a fresh-burst fast-path that scans raw mentions (ticker's first day with a burst), bypassing the 20-mention gate — for LIVE catching, not just post-hoc.
- **Calendar effects (10y, SPY/QQQ/Gold; absent in AUDJPY-FX)**: turn-of-month is real (SPY +0.49 vs +0.20%/5d, QQQ +0.58 vs +0.35, Gold +0.38 vs +0.25); September weak (-0.2%); OpEx week underperforms. Generic equity-flow seasonality → useful as *overlays/filters*, not Reddit alpha.
- **AU sentiment track** building ( `au_aggregate_daily`, post-level) to test AU-sentiment → AUDJPY.



## ★ MASTER BACKTEST vs BUY-AND-HOLD ( `backtest/run_all.py` , full 2020-2026, 10d, net of costs)

Every signal judged on EXCESS over its instrument's unconditional drift (buy-and-hold-anytime). | Strategy | n | net% | B&H% | excess | win | verdict | |---|---|---|---|---|---| | retail\_fear AUDJPY | 338 | +0.53 | +0.27 | **+0.25** | 69% | beats (modest, most robust) | | retail\_fear SPY | 338 | +0.54 | +0.59 | -0.05 | 61% | **no edge** | | retail\_fear Gold | 338 | +0.83 | +0.72 | +0.10 | 57% | marginal | | euphoria\_short SPY | 351 | -1.26 | +0.59 | -1.84 | 27% | no edge | | smallcap\_pump\_fade | 87 | +9.07 | +0.59 | +8.48 | 64% | beats but single-regime + brutal costs | | biotech\_catalyst | 91 | +22.2 | +0.59 | +21.6 | 36% | fat-tailed LOTTERY (low win) | | attention\_fade | 252 | -0.73 | +0.59 | -1.31 | 40% | no edge |

**Recalibration:** with a proper buy-and-hold benchmark over the full period, the AUDJPY edge is a **modest +0.25pp/10d** (earlier raw "positive every year" overstated it - that was largely drift). SPY capitulation does NOT beat buy-and-hold. The big-excess equity signals are single-regime (smallcap) or low-win lotteries (biotech). Net: one modest-but-real macro edge, several promising- but-unvalidated screens. Do NOT size up on these as-is.

## DATA WE NEED (to validate/strengthen — the binding constraints)

| Data                                                                        | Unblocks                                                                                                                    | Why we lack it                                                                                                           |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Intraday prices (we have EOD only)                                          | pump entry/exit timing; fresh-pump fade execution (pumps move intraday)                                                     | not collected                                                                                                            |
| Historical penny-stock prices + per-ticker features                         | multi-regime validation of smallcap_pump_fade & biotech (only 2025-26 exists)                                               | delisted pennies → Yahoo gaps; per-ticker feature_daily only recent                                                      |
| Short interest / float / borrow fee                                         | squeeze setups; realistic short cost/feasibility                                                                            | FINRA daily short-volume in ( <code>scraper/finra.py</code> ); still need SI%/float/borrow (Ortex / FINRA bi-monthly SI) |
| Options chains / IV                                                         | express fades via PUTS (only sane way to short pennies); options-flow signal                                                | not collected                                                                                                            |
| More years of RRAI history                                                  | tighten the macro edge (effective n ≈ a few dozen fear episodes)                                                            | Reddit/arctic-shift depth ~2017+                                                                                         |
| Broader author coverage (~20% profiled)                                     | a PIT-tradeable deletion/young-account short (thesis already validated forensically; <code>young_frac</code> proxy starved) | rate-limited Reddit /about sweep                                                                                         |
| Independent sentiment sources (StockTwits, news, Discord, crypto + AU subs) | independent episodes → real effective-sample growth without waiting calendar time                                           | not integrated (AU track building)                                                                                       |
| Real transaction-cost data (per-instrument spreads/borrow)                  | trustworthy net returns (small-cap costs are modelled/optimistic)                                                           | not sourced                                                                                                              |

## Update — complete-data confirmation (2026-06-01)

- US backfill finished: continuous 2020-2026 (2,344 days, 341 capitulation days). **Capitulation-long multi-regime result HOLDS on the complete series** — AUDJPY positive every year incl. 2022 bear (win 60-83%).
- `smallcap_pump_fade` is strong (+12.4% excess short, win 77%) but **only 2025-26 data exists** (n=87) — no historical penny-stock prices/features, so it **can't be multi-regime validated** → stays ADVISORY, NOT promoted to tradeable. (Single-regime; can't be fixed by backfill — penny price history doesn't exist.)

## Crypto — coincident, NOT a leading indicator (tested 2026-06-03)

Added CryptoCurrency/Bitcoin/CryptoMoonShots/SatoshiStreetBets (~1yr; Bitcoin 61k posts) and tested whether crypto leads equities / risk-appetite. **It doesn't — it's coincident:** - **Price lead-lag (BTC↔SPY, 2,577d):** same-day corr **+0.17 (t=8.8)** but every

lead/lag  $k \neq 0$  is noise ( $|t| < 1.3$ ) — neither leads. - **Weekend test** (crypto's best shot — BTC trades Sat/Sun while equities are closed; 490 weekends): corr +0.05,  $t=1.2$  — no lead. - **Crypto sentiment** → **equity RRAI**: corr **peaks at  $k=0$  (+0.20,  $t=3.9$ )** then decays (a lead would peak at  $k>0$ ). Barely predicts even BTC's *own* next-day return (+0.05). - **Combined crypto+equity RRAI capitulation (1yr overlap,  $n=55$ )**: does **NOT** help — for the validated AUDJPY edge it *dilutes* it (equity-only excess +0.55% > combined +0.45% > crypto-only -0.03%). Crypto sentiment is redundant with (and noisier than) the equity RRAI. - One weak whiff: crypto *euphoria* → mildly negative SPY 3d ( $t=-2.4$ , but overlapping windows → true  $t \approx -1.4$ ) — *contrarian*, not leading. Watch, not an edge.

**Takeaway:** crypto's value is NOT macro timing (it's a redundant coincident risk-appetite read). It's worth keeping only as its own **pump/meme universe** (CryptoMoonShots/SatoshiStreetBets) for the `distribution_short` /pump signals. Do not fold it into the RRAI.

## Changelog

- **2026-06-03 — Crypto leading-indicator test: COINCIDENT, not leading.** Price lead-lag  $BTC \rightarrow SPY \approx 0$  (same-day +0.17 but  $k \neq 0$  noise), weekend test  $t=1.2$ , crypto sentiment coincident with equity RRAI (peaks  $k=0$ ). Combined crypto+equity RRAI does NOT improve capitulation — *dilutes* the AUDJPY edge (+0.55% → +0.45%). Crypto's value = its own pump/meme universe, not macro timing. (See "Crypto — coincident" section.)
- **2026-06-03 — RRAI extended to 2016-06 (525 caps, +190) → fear-gate STRENGTHENED.** Backfill landed (aggregate\_daily 2016-06 → 2026, 3,653 days; +2018 Volmageddon/Q4 episodes). Re-ran `backtest.fear_gate`: fear-only excess grew SPY +0.32 → +0.57, QQQ +0.41 → +0.80 (effect *grew* OOS = real-edge signature); episode-level now **significant on BOTH legs and they agree** (SPY  $t=2.77$ , QQQ  $t=2.37$ ,  $k=63$ ) — the prior SPY ≠ QQQ noise-floor flag is resolved. Day-level still marginal (1.6-1.9). The LIVE `trend+fear_gate` is still  $\approx 0$  (SPY +0.10/QQQ +0.11) ⇒ **drop the trend filter, use fear-only**. AUDJPY confirmed best ungated.
- **2026-06-02 — Power analysis + RRAI history extension (in progress).** Quantified the fear-gate's data need: the 94 fear-days are ~40 independent episodes; day-level it's  $t \approx 0.6$  (needs ~9× more obs for  $p < 0.05$ , ~19× for power — impractical), but episode-level it's  $t=2.28$  (SPY)/1.85 (QQQ) — significant-but-fragile (COVID-cluster-driven; SPY ≠ QQQ ⇒ noise floor). Cheapest lever = more *independent fear episodes* via **backward** history (Wikipedia fear data already reaches 2015). Extended study prices to max and **kicked off the RRAI aggregate backfill 2016-06 → 2020** (wallstreetbets/stocks/pennystocks) to add the 2018 Volmageddon + Q4-2018 episodes; will re-run `backtest.fear_gate` on the longer series when it lands.
- **2026-06-02 — Committed EXCESS study settles the fear-gate ( `backtest/fear_gate.py` ).** The proper frame (mean excess vs buy-&-hold, the +0.36pp's own metric). Three findings: (1) the +0.36pp **reproduces** as a point estimate (SPY +0.32 / QQQ +0.41 fear-only excess) — not fabricated; (2) it's **not significant** (excess  $t \approx 0.6$ ,  $n \approx 94$ ); (3) the LIVE **trend+fear** combo is *worse* — dilutes SPY (→ +0.09), flips QQQ negative (→ -0.20) — because the trend filter strips the downtrend bounces that carry the excess. Reconciles the MT5 "loss": MT5 sums absolute P&L (COVID-cluster dominates), excess means-averages — so the earlier "drag" was a frame artifact. Gate stays 🟡; fear-only > trend+fear for excess; AUDJPY stays ungated (the only significant edge). Exporter refactored to share the study's `capitulation_sets()` (MT5 + Python test identical dates).
- **2026-06-02 — MT5 backtest of the fear-gate → it's a DRAG, not a rescue.** Extended `--export-signals` to emit 4 point-in-time capitulation date-sets (baseline / `fear_z ≥ 0.5` / 200-DMA-trend / trend+fear) and ran each through the MT5 Strategy Tester on US500 (2020-26) via `RedditMacro_US500*.ini`. Adding `fear_z` made absolute returns *worse* (alone -\$1045, with trend -\$372 vs trend-only +\$21) — fear spikes fire long into tops (Feb-2020). Only the trend filter helps (DD 13.6 → 7.1%). Caveat: MT5 = absolute long-only, the +0.36pp was *excess* — so not a direct refutation, but the gate is downgraded to 🟡 doubtful and needs a committed *excess-based* study to settle (maybe drop `fear_z`, keep trend).
- **2026-06-02 — Doc-completeness pass.** Audited the doc against the live registry + all backtest modules. Found one missing edge — the **Wikipedia fear-confirmation gate** (live: gates the equity legs of `retail_fear`, `fear_z ≥ 0.5`, +0.36pp rescue) — now documented, *honestly flagged as not yet committed-backtested*. Also added the full live-screen list ( `fresh_pump_alert`, `coordination_flag`, ...) to the scoreboard and recorded the broad "attention-leads-price" long as a failed up-leg test.
- **2026-06-02 — New PIT-tradeable short found: `distribution_short` ( `backtest/microstructure.py` ).** Built the missing **new-author recruitment** feature and tested demand-quality timing signals. The decisive lesson: microstructure timing (recruitment-exhaustion / deceleration / divergence) is **worthless on the broad attention universe** (it drifts flat-to-up, costs dominate) but **powerful inside the concentrated/manufactured subset** — exactly where the deletion edge lives. Winner = concentrated pump + euphoric sentiment + price already rolling over → **+7.2%/10d net,  $t=2.70$  (date-clustered  $t=2.52$ , 47 indep. dates), dose-response, survives 2× costs**; control loses -4.1%. Unlike the deletion gate it uses **no hindsight** → tradeable. Recorded negatives (broad timing, recruitment-alone, up-leg ride) to avoid re-runs.
- **2026-06-02 — Deletion-gated short forensically validated.** Author coverage 8% → 20% (2,973 gone). Re-backtest: `gone_frac ≥ 0.20` → +14.3%/10d short (83% win,  $t=2.57$ , +ve all regimes); the *control* (un-gated concentrated pumps) *loses* -10.8% — so the deletion gate flips the trade, confirming the A2 thesis. **Caveat:** `gone_frac` is hindsight (no Reddit deletion

timestamp) → not yet PIT-tradeable; the `young_frac` PIT proxy is still too sparse (n=3–7). Also shipped `scraper/finra.py` (FINRA Reg SHO daily short-volume, 475k rows) + z-score `squeeze_screen()` — learned the *absolute* short-ratio is market-maker noise (median 0.49), so the screen uses per-ticker z-score anomaly + chatter.

- **2026-06-01** — Instrument scan (17 MT5 instruments). `retail_fear` generalises across the risk-on basket; commodities express it best (Silver +2.2pp lumpy, Gold +0.6pp robust), AUDJPY steadiest. Oil/crypto dropped. Basket = AUDJPY + Gold + indices + small Silver; EA + feed updated.
- **2026-06-01** — Macro R&D + MT5 framework. Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals` feed; retired `RedditRRAI_EA.mq5`.
- **2026-06-01** — Initial document. Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.